EXPERIMENT 1

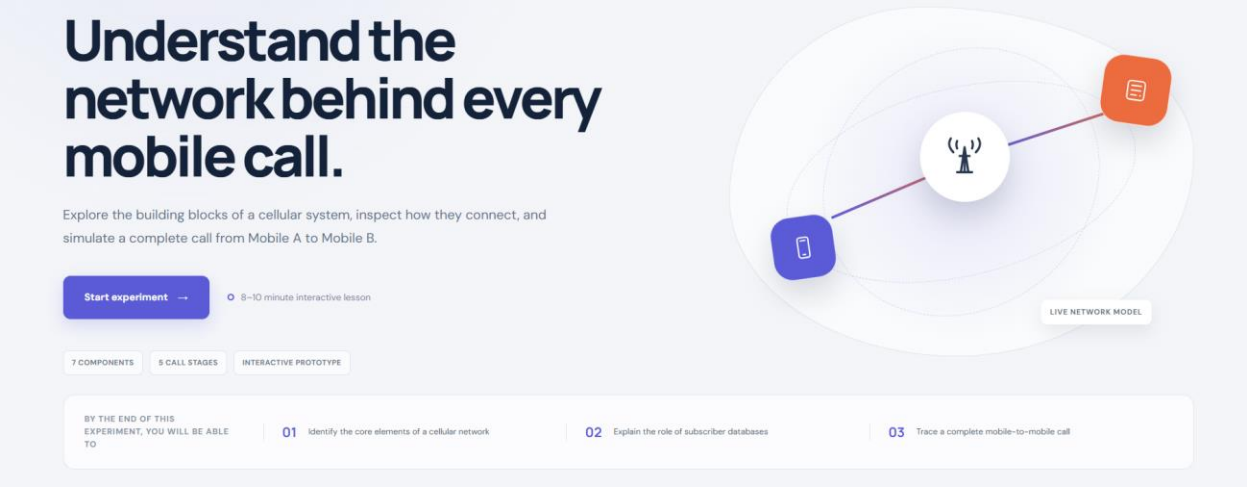
AIM

To design an interactive diagram using Figma illustrating the components of a basic cellular system.

PROCEDURE

1. Open Figma and create a new design file.
2. Create shapes and labels to represent the main components of a cellular system, such as the Mobile Station (MS), Base Transceiver Station (BTS), Base Station Controller (BSC), Mobile Switching Centre (MSC), and Public Switched Telephone Network (PSTN).
3. Arrange the components in a logical sequence and connect them using lines and arrows to show communication flow.
4. Add colors, icons, and text labels to make the diagram clear and attractive.
5. Use Figma's prototyping features to add interactive elements, such as clickable components and navigation between frames.
6. Preview the prototype to verify that all interactive elements work correctly.

DESIGN



One network, three functional layers

A cellular system combines radio access, switching, and subscriber data to keep users connected while they move.

01

Radio access

The mobile station and BTS create the wireless link into the network.

AIR INTERFACE

02

Control & switching

The BSC manages radio resources while the MSC routes and controls calls.

NETWORK CORE

03

Subscriber data

HLR and VLR records help authenticate subscribers and locate their devices.

DATA SERVICES

THE BIG IDEA

Radio gets you in. The core gets you connected.

External calls can leave the mobile core through the PSTN, while mobile-to-mobile calls remain within the cellular network.

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Continue →

Meet the network components

Select a card to inspect its role and key responsibilities.

7 ESSENTIAL COMPONENTS

MS
Mobile Station →

BTS
Base Transceiver Station →

BSC
Base Station Controller →

MSC
Mobile Switching Center →

HLR
Home Location Register →

VLR
Visitor Location Register →

PSTN
Public Switched Telephone Network →

MS · SELECTED COMPONENT

Mobile Station

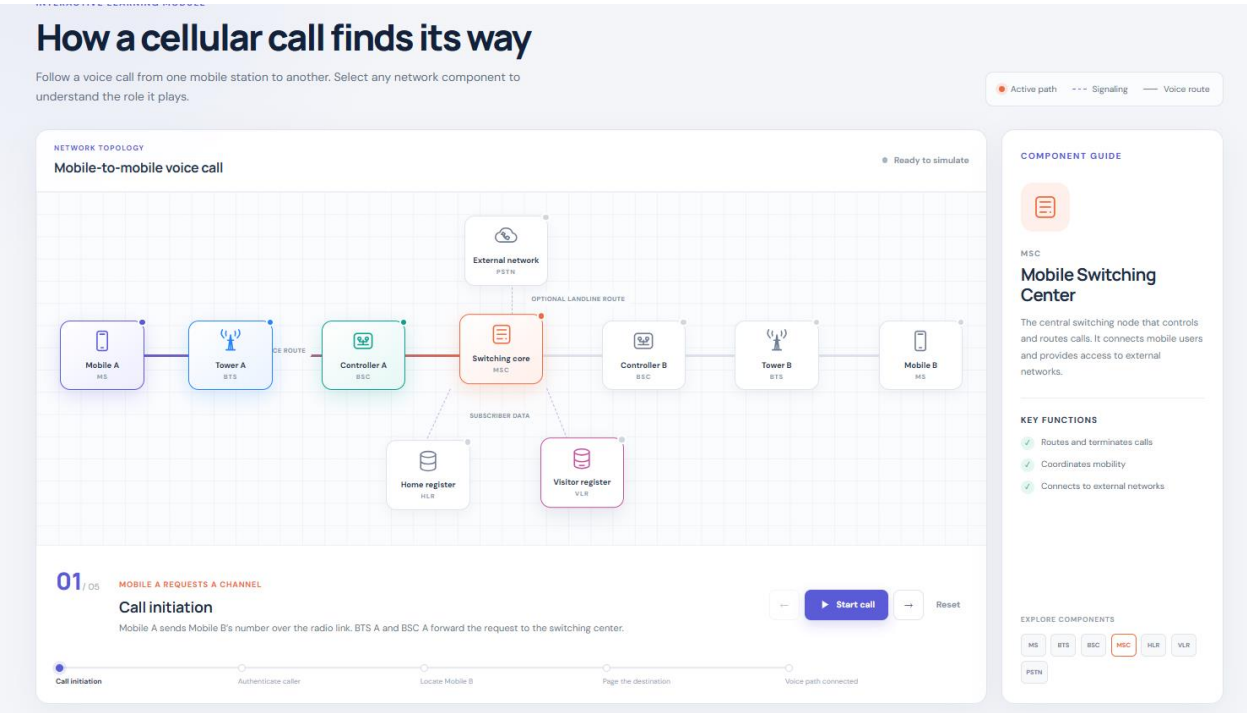
The user's device and subscriber identity. It converts voice into radio signals and connects to the nearest cell tower.

KEY FUNCTIONS

- ✓ Starts and receives calls
- ✓ Carries subscriber identity
- ✓ Measures nearby signal strength

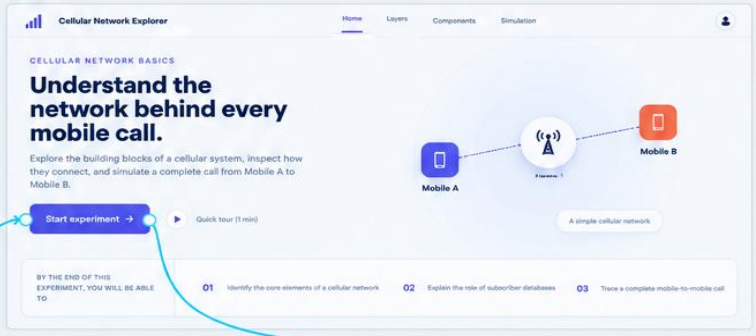
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Open simulation →

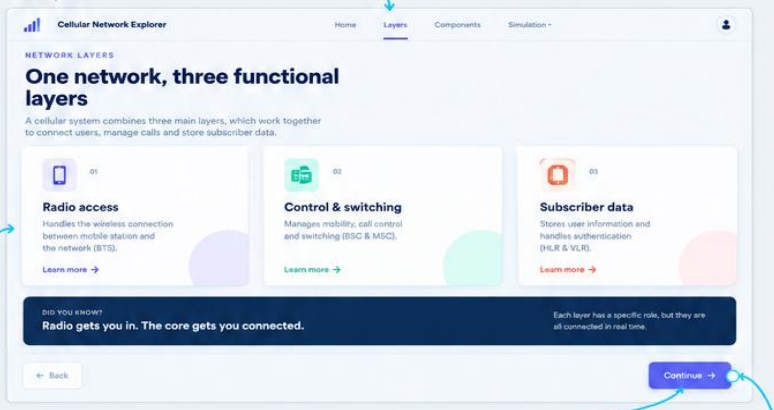

How a cellular call finds its way
Follow a voice call from one mobile station to another. Select any network component to understand the role it plays.
Active path
Signaling
Voice route
NETWORK TOPOLOGY
Mobile-to-mobile voice call
Ready to simulate
External network
PSTN
OPTIONAL LANDLINE ROUTE
Mobile A
MS
Tower A
BTS
Controller A
BSC
Switching core
MSC
Controller B
BSC
Tower B
BTS
Mobile B
MS
SUBSCRIBER DATA
Home register
HLR
Visitor register
VLR
01/05
MOBILE A REQUESTS A CHANNEL
Call initiation
Mobile A sends Mobile B's number over the radio link. BTS A and BSC A forward the request to the switching center.
Start call
Reset
Call initiation
Authenticate caller
Locate Mobile B
Page the destination
Voice path connected
COMPONENT GUIDE
MSC
Mobile Switching Center
The central switching node that controls and routes calls. It connects mobile users and provides access to external networks.
KEY FUNCTIONS
Routes and terminates calls
Coordinates mobility
Connects to external networks
EXPLORE COMPONENTS
MS
BTS
BSC
MSC
HLR
VLR
PSTN

PROTOTYPE

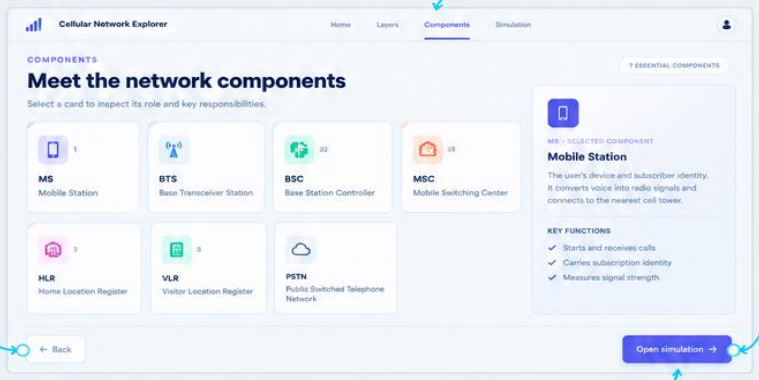
Desktop - 1



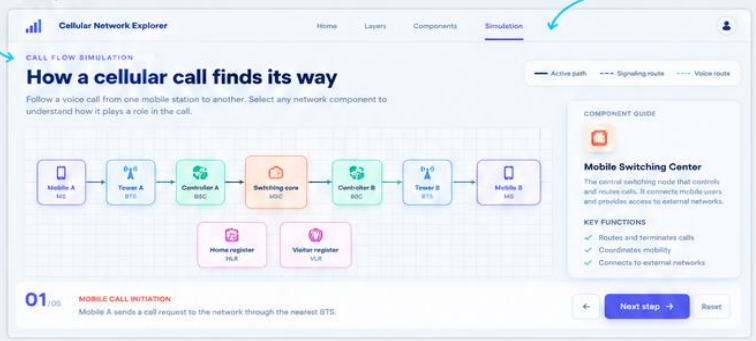
Desktop - 2



Desktop - 3



Desktop - 4



RESULT

Thus, an interactive diagram illustrating the components of a basic cellular system was designed using Figma, showing the major components and their communication flow.